Computer COMP-111

Important Questions

Short Questions

- Different b/w Web page & Web browser?
- Different b/w Intranet & Internet.

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- Different b/w Web page & Web browser?
- Different b/w Intranet & Internet.

- Define Database & its advantages?
- Define Modulation & Demodulation?
- Different b/w LAN & WAN?

- Different b/w Singletasking & Multitasking.
- Define World Wide Web (www)?
- Define Network & Protocol?
- Different b/w CLI and GUI?

- Different b/w LCD & LED?
- Define Volatile & Non-Volatile memory?
- Different b/w SSD and HDD?
- Define E-commerce?
- Define Search engine & URL?
- Different b/w CD and DVD?
- Define Computer network with types?
- Different b/w RAM & ROM?
- Different b/w PROM, EPROM and EEPROM?
- Magnetic disk & Hard disk?
- Different b/w Compiler & Interpreter?

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- Different b/w RAM & ROM? BS Notes Bank
- Different b/w PROM, EPROM and EEPROM?
- Magnetic disk & Hard disk?
- Different b/w Compiler & Interpreter?
- Define Data communication & Data transmission?
- Use of computer
- Computer Virus
- Data & Information.

Long Questions

- Components of Computer system
- Types of Computer?
- Components of System Unit?
  - Motherboard • CPU • Memory or Storage Unit
  - Control unit • Arithmetic Logic Unit (ALU)
- Computer Bus
- Input Devices (briefly) BS Notes Bank
- Output Devices (briefly)
- Biometric Input devices
- Scanner with types
- Communication media with types

Solved Past Paper

1. Describe the function of Arithmetic and Logic Unit?

An ALU is that part of a computer's CPU which performs arithmetic and logic calculations.

Function of ALU:-

The function of an ALU is to take binary inputs, execute the operation, and create, store, and distribute binary output. ALUs make arithmetic

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2) RAM is called volatile memory. Justify.

RAM is volatile memory, which means that the information temporarily stored in the module is erased when you restart or shut down your computer. Because the information is stored electrically on transistors so when there is no electric current, the data disappears.

3) State the use of copy paste function in MS word.

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When you copy text or an image, you are placing a duplicate in the Clipboard, but you are not removing it from its original location. After cutting or

copying, you can then use the paste function to

copying, you can then use the paste function to move the text or image from the clipboard. to a new location in your document.

4) What is EEPROM?

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EEPROM is a type of non-volatile ROM that enables individual bytes of data to be erased and reprogrammed. EEPROM is usoally used to store small amounts of data in computing and other electronic devices.

6) What are super computers?

Super computer

Supercomputers are the largest, fastest, most powerful, and most expensive computers made. Supercomputers can be accessed by many individuals at the same.
The first supercomputer was built in the 1960s for the United States department of Defense.

Uses in daily life:-

The first supercomputer was built in the 1960s for the United States department of Defense.
Uses in daily life:-

- Used for weather prediction.
- Used for weapon design and atomic research.
- Government use it for different calculations and heavy jobs.

7) Difference b/w system software and application system.

System Software	Application Software
• System software are mainly designed for managing system resources.	• Application Software are designed to accomplish tasks for specific purposes
• Programming of system software is complex.	• Programming of application software is comparatively easy.

A computer cannot run without system software.	A computer can easily run without application software.
System software do not depend on application software	Application software depend on system software and cannot run without system software.

8) Explain Modes of Data Transmission?

Transmission mode means transferring of data between two devices. It is also known as communication mode.

There are three types of transmission modes:

- Simplex Mode
- Half - Duplex Mode
- Full- Duplex Mode

What are Nested Functions in Excel?

Nested IF Function

The IF function in Excel can be nested. The Nested IF function is a combination of multiple IF functions. A 'nested IF' contains at least one IF function inside another to evaluate more than one condition at once and return the output accordingly. In Excel, up to 64 IF functions can be nested in a formula. But, we must double-check to ensure that each IF condition

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is correctly nested inside the other.

10) Differentiate b/w Analog and Digital Computers?

1.9.4 Difference between Analog and Digital

BASIS FOR COMPARISON	ANALOG COMPUTERS	DIGITAL COMPUTERS
Basic	Computations are carried out employing continuous variations of physical properties.	Discrete electrical voltage levels are used to encode the input.
Computation	Performed in real-time	Large delays can occur
Accuracy	Low	High
Circuitry	Made up of electrical and mechanical components.	Consists of a large number of tiny electrical switches.
Mechanism	Uses variation of different physical quantities.	Uses binary numbers and binary arithmetic.
Examples	Speedometers, energy meters and	Digital cameras and

	continuous variations of physical properties.	levels are used to encode the input.
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Circuitry	Made up of electrical and mechanical components.	Consists of a large number of tiny electrical switches.
Mechanism	Uses variation of different physical quantities.	Uses binary numbers and binary arithmetic.
Examples	Speedometers, energy meters and traditional washing machine, etc	Digital cameras and watches, thermometers, scanners, modern computers, etc.

LONG QUESTIONS

1) Explain different types of Non-Impact Printers?

3.18.3 Non-Impact Printer

A type of printer that does not operate by striking a head against a ribbon. Examples of nonimpact printers include laser and ink-jet printers. The term non-impact is important primarily in that it distinguishes quiet printers from noisy (impact) printers. Non-Impact printers are faster and produce high quality output than impact printers. They can print up to 24 pages per minute. They produce no noise during printing. These printers are costly than impact printers.

The Examples of Non-Impact printers are:

1- Laser Printer
2- Inkjet Printer
3- Thermal Printer

Laser Printer

Laser stands for Light Amplification by Simulated Emission of Radiation. A laser printer is the fastest and high quality non-impact printer. It works like a photocopier. The laser printer

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transfers the image of output on paper using LASER technology and toner. Toner is an ink powder. It is used in laser printers and photocopiers also. The laser printer has a special drum inside it. First the image of output is created on the drum, and then it is transferred from drum to paper. The image of output is created on the drum by throwing magnetic ink powder in the form of microscopic dots. These dots can be from 300 dpi to 1200 dpi (dpi means dots per

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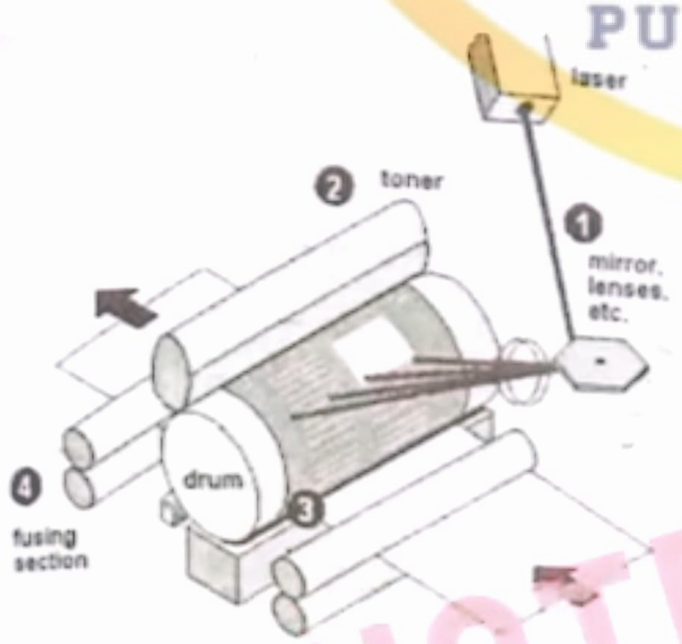
The Laser printer can print both text and graphics in very high quality resolution. Laser printer prints one page at a time. The laser printers are, therefore also called page printers. The printing speed of laser printer is about 4 to 32 pages per minute for microcomputer and up to 200 pages per minute for mainframe computers.



Ink Jet Printer

Ink-jet printer is type of non-impact printer. It creates output on paper by spraying tiny drops of liquid ink. Inkjet printer has print-head that can spray very fine drops of ink. It consists of print cartridge filled with liquid ink and has small nozzles in form of matrix. Like dot cartridge filled with liquid ink and has small nozzles in form of matrix. Like dot matrix printer, the combination of nozzles is activated to form the shaper of character

or image on the paper by spraying the liquid ink. These printers have resolution ranging fr 300 to 720 dpi.

The ink-jet printers have low price than laser printers. They are also slower and have low quality than laser printer. However, they are faster and have high print quality than dot ma printers. The printing speed of ink-jet printer is from 1 to 6 pages per minute.

Thermal Printer

Thermal printer is another type of non-impact printer. It can only print output on a special sensitive waxy paper. The image if the output is created on the waxy paper by burning dot it. For colored output, colored waxy sheets are used. Thermal Printer produces a high qual printout. It is quite expansive as compared to other non-impact printers.

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2) Define Storage Devices, explain all major types of storage devices.

4.1 Storage Devices

A storage device is a piece of computer hardware used for saving, carrying and pulling out data. It can keep and retain information short-term or long-term. It can be a device inside or outside a computer or server. Other terms for storage device is storage medium or storage media.

A storage device is one of the basic elements of any computer device. It almost saves all data and applications in a computer except for hardware firmware. It comes in different shapes and sizes depending on the needs and functionalities.

There are two types of storage devices used with computers: a primary storage device, such as RAM, and a secondary storage device, such as a hard drive. Secondary storage can be removable, internal, or external.

4.3 Hard Disk

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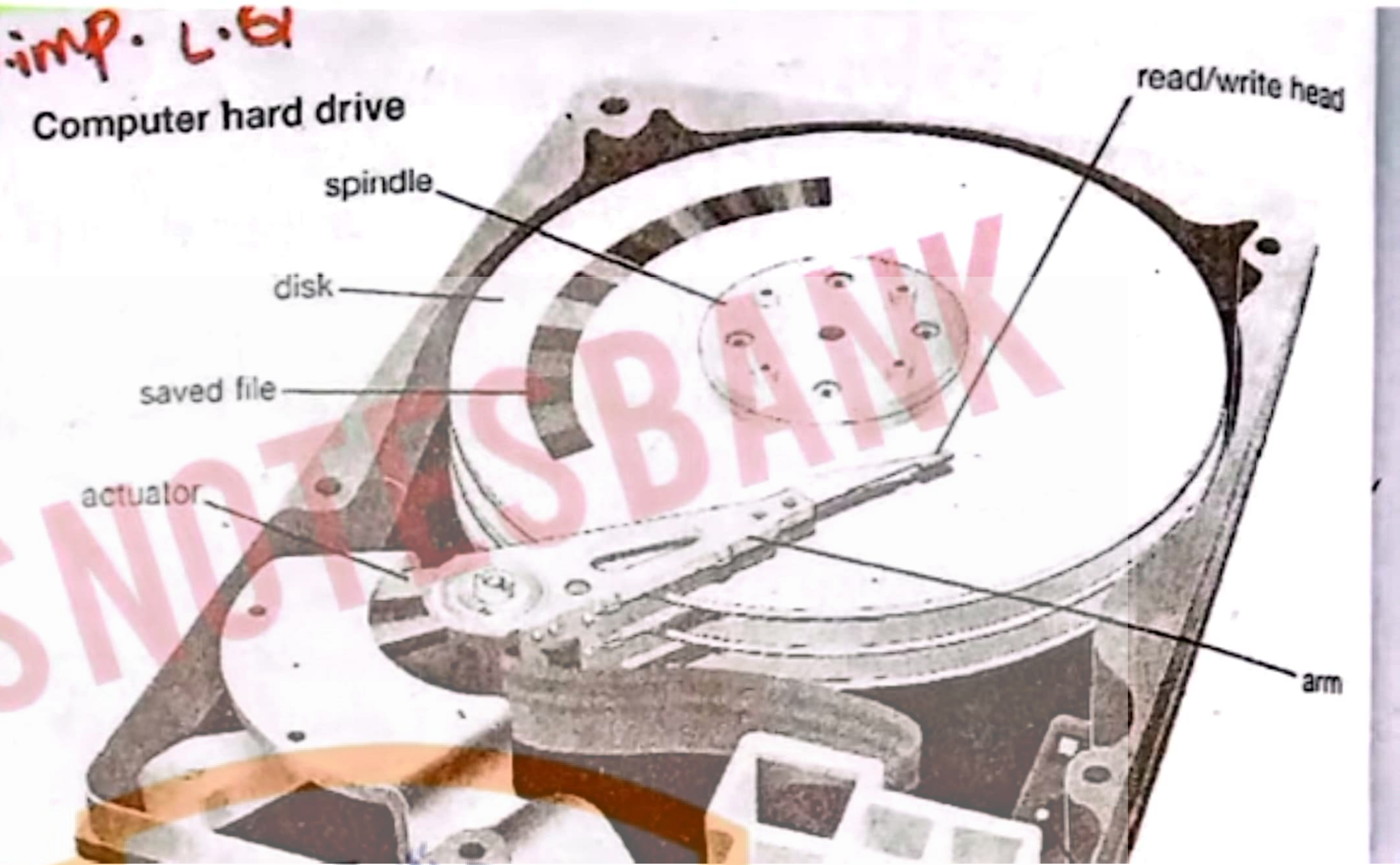
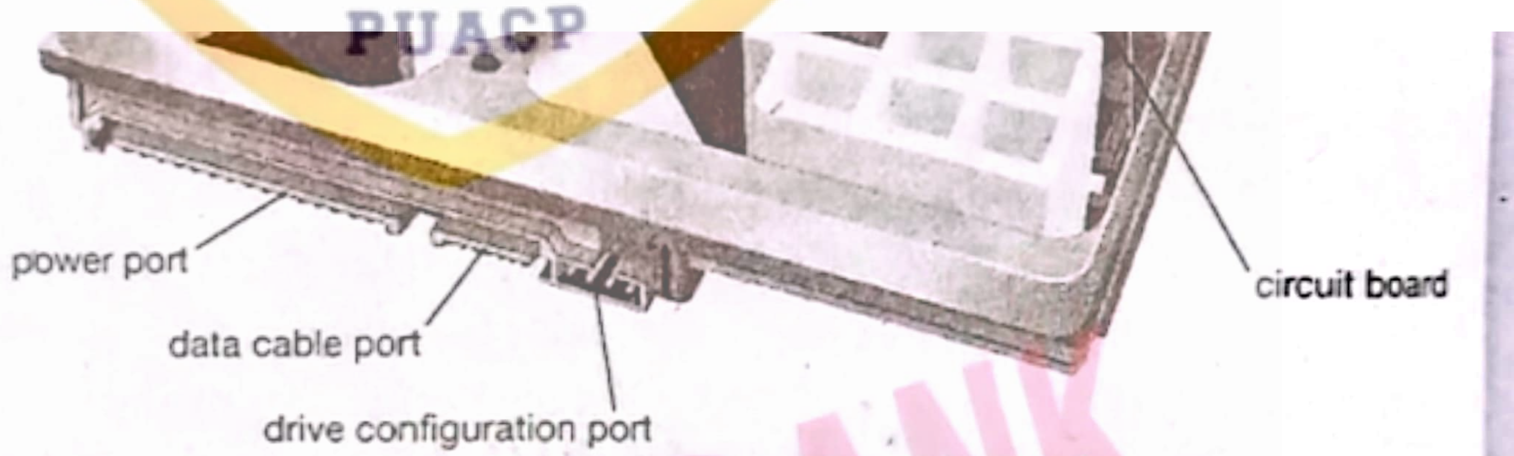
When you save data or install programs on your computer, the information is typically written to your hard disk. The hard disk is a spindle of magnetic disks, called platters, that record and store information.



Because the data is

Because the data is
stored magnetically, information recorded to the hard disk remains intact after you turn your computer off. This is an important distinction between the hard disk and RAM, or memory, which is reset when the computer's power is turned off.



The hard disk is housed inside the hard drive, which reads and writes data to the disk. The hard drive also transmits data back and forth between the CPU and the disk. When you save data on your hard disk, the hard drive has to write thousands, if not millions, of ones and zeros to the hard disk. It is an amazing process to think about, but may also be a good incentive to keep a backup of your data. Formatting is a process that creates tracks and sectors on the disk. Tracks are in the form of circles on the surface of a hard disk. Each track on a disk is divided computer off. This is an important distinction between the hard disk and RAM, or memory, which is reset when the computer's power is turned off.

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4.3.1 Hard Disk Performance

Seek Time: It is also positioning performance. It is the time required by read/write head to reach the correct location on the disk. It I often used with rotational speed to compare performance of hard drives. It is measured in milliseconds (ms).

Spindle speed: It is also called transfer performance. It is the speed at which the drive transfer data. It is measured in revolutions per minute (RPM).

Latency: It is the time required by the spinning platter to bring the desired data to read/write head. It is measured in milliseconds (ms).

4.3.2 External and Removable Hard Disks

An external hard drive is a storage device located outside of a computer that is connected through a USB cable or wireless connection. An external hard drive is usually used to store media that a user needs to be portable, for backups, and when the internal drive of the computer is already at its full memory capacity. These devices have a high storage capacity compared to flash drives and are mostly used for backing up numerous computer files or serving as a network drive to store shared content. External hard drives are also known as removable hard drives.

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A type of disk drive system in which hard disks are enclosed in plastic or metal cartridges so that they can be removed like floppy disks. Removable disk drives combine the best aspects of hard and floppy disks. They are nearly as capacious and fast as hard disks and have the portability of floppy disks. Their biggest drawback is that they're relatively expensive.

Advantages of External Removable Hard Disks

Additional Storage

Portable hard drives can provide much more space, freeing up room on the main computing system. With capacities of 80 GB to over 1 terabyte, external hard drives allow an even greater storage space for digital media such as photos, music and home movies. You can store entire virtual workstations on portable hard drives.

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Portability

Often no larger than the size of a child's shoebox, all external hard drives are portable to a certain extent. Portable hard drives allow users to secure data in a safe place, share it with other users or transfer files to another computer with very simple plug-and-play interface.

Plug-And-Play

Portable hard drives operate on a plug-and-play basis. Many portable hard drives come with both USB and Fire-wire, which allows for universal connectivity for both PC and Mac. Any system with this capability will recognize the portable hard drive automatically and assign it a letter to designate it. Be sure to check for this capability.

Backing Up Important Data

A portable hard drive allows users to back up all their files. If a computer needs to have its hard drive reformatted, a portable hard drive has much more efficient capability for backing up files as opposed to lower capacity DVDs or CD-Rs. Many of these drives come with

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4.4 Floppy Disk or Diskette

Floppy disk is also called diskette. It consist of a thin plastic disk coated with magnetic material. This is enclosed in a plastic jacket. It was introduced by IBM in early 1970s. It is a portable storage medium and san be removed from one computer and inserted into another computer easily. It is not commonly used now a days.

Floppy disk can only store a small amount of data. Data access speed of floppy disk is slower than hard disk. It is inexpensive storage media. The standard size of floppy disk is 3 ½ inch. The capacity of floppy disks is 1.44 MB. The circular piece of plastic on 3 ½ inch diskette is enclosed in a shell. A piece of metal covers the reading and writing area. It is called shutter. When the disk is inserted into a disk drive, the shutter opens to expose the surface of the disk.

Density: It is the number of bits that can be stored in one inch. It is measured as bits per inch (bpi). A floppy disk with higher density stores more data.

Number of Tracks: The number of tracks on a diskette is measured ...

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Density: It is the number of bits that can be stored in one inch. It is measured as bits per inch (bpi). A floppy disk with higher density stores more data.

Number of Tracks: The number of tracks on a diskette is measured as tracks per inch (tpi). A floppy disk with more tracks can store more data.

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4.7 Flash Memory Storage

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Flash memory is a non-volatile memory chip used for storage and for transferring data between a personal computer (PC) and digital devices. It has the ability to be electronically reprogrammed and erased. It is often found in USB flash drives, MP3 players, digital cameras and solid-state drives.

Flash memory is a type of electronically erasable programmable read only memory (EEPROM), but may also be a standalone memory storage device such as a USB drive.

EEPROM is a type of data memory device using an electronic device to erase or write digital data. Flash memory is a distinct type of EEPROM, which is programmed and erased in large blocks.

4.7.1 Solid State Drives (SSD)

Short for solid-state drive, an SSD is a storage medium that uses non-volatile memory to hold

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data. Flash memory is a distinct type of EEPROM, which is programmed and erased in large blocks.

Types

4.7.1 Solid State Drives (SSD)

Short for solid-state drive, an SSD is a storage medium that uses non-volatile memory to hold and access data. Unlike a hard drive, an SSD has no moving parts, which gives it advantages, such as faster access time, noiseless operation, higher reliability, and lower power consumption. The picture shows an example of an SSD made by Crucial.

As the costs have come down, SSDs have become suitable replacements for a standard hard drive in both desktop and laptop computers. SSDs are also a great solution for netbooks, nettops, and other applications that don't require a lot of storage.

3= They are light-weight.

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3) Explain how you create and manipulate charts in MS Excel. Also Explain Slide Animations and Animation Effects of MS Powerpoint?

Check the Description Box

5) What is modem? also name its types.

The full form of modem stands for modulator-demodulator. A modem is a hardware that connects to a computer, broadband network or wireless router. Modem converts information between analogue and digital formats in real time making seamless two-way network communication.

Types of Modems

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Types of Modems

There are three types of modems:

1) Landline modems
2) Wireless modems
3) LAN modems

MS-Word

Microsoft Word or MS Word (often called Word) is word processing software that allows users to create and edit text documents.

→ Word processing documents include:

- Letters
- Memorandums
- Faxes
- Mail Merges
- Reports
- One page flyers
- E-mail

- **History of Microsoft Word:**

The first version of Microsoft Word was developed by Charles Simonyi Richard Brodie, former Xerox programmers hired by Bill Gates and Paul Allen in 1981.

Microsoft Word, first released in 1983 as "Multi-Tool Word," is a word processor available as a standalone product and as a component in the Microsoft Office suite. The first version of Microsoft Word was based on the framework of Bravo, the world's first word processor with a graphical user interface.

Features of MS Word

- MS Word offers easy editing of text. Using word you can create the document and edit them later as and when required, by adding more text, modifying the existing text, deleting/moving some part of it.

- MS Word uses WYSIWYG (What you see is what you get display.
- Front size and type of font can also be changed, page number and header and footer can be included.

- Saving and retrieving of documents
- Cut and paste feature
- Document formatting
- Printing facility
- Spell check
- Mail merging
- Headers and footer
- Bold, italics, underline.
- Find and replace
- Rulers
- Bullets and numbering
- Support for tables
- Support for macros
- Support for multiple languages.

Uses of MS Word in daily life

- Bussiness and Workplaces use MS Word to create Emails, Bills, Cash Memo, Joining letters, and so on.

- In Education sectors MS Word is used to create study papers, lectures or Journals. Students use MS Word for completing their homeworks or writing their assignments.
- Home based uses of Microsoft Word includes creating a birthday card, invitation card or to write a letter or dairy.

- MS Word helps you to get a job.

- It helps in creating resumes, notes and assignments.

- MS Word is used to create document and convert it into PDF document.
- MS Word is used to collaborate with team members.
- You can use MS Word to teach others.